

Q1. Write a Python program to create a DataFrame containing the attributes Age, Salary, Experience, Working_Hours, and Performance_Score. Apply the Standardization data transformation technique and display the transformed data.

[10 Marks]

Q2. Create a dataset containing information about different books with the attributes Book Title, Author, Publication Year, Average Rating, and Number of Ratings.

Perform the following tasks on created dataset:

- Create a Venn Diagram comparing the authors of books published in two different time periods.
- Generate a Word Cloud using the book titles.

[20 Marks]

OR

Q2. Apply the Apriori algorithm to the following student activity dataset to identify frequent itemsets and generate association rules among the activities. Repeat the analysis using minimum support values 0.2 and 0.3.

[20 Marks]

TID	Items
1.	Library , Seminar
2.	Workshop, Project
3.	Library, Assignment
4.	Seminar, Workshop
5.	Assignment, Seminar
6.	Workshop, Assignment
7.	Project, Library
8.	Library, Seminar, Project
9.	Library,Seminar,Workshop

Q3. Viva

[5 Marks]